AGENTREEL PROCESS REPORT: FROM PROMPT TO PRODUCT

How 7 Autonomous AI Agents Build a Broadcast-Ready Vertical Video

Organisation: LY Agentic AI Laboratory, K J Somaiya School of Engineering

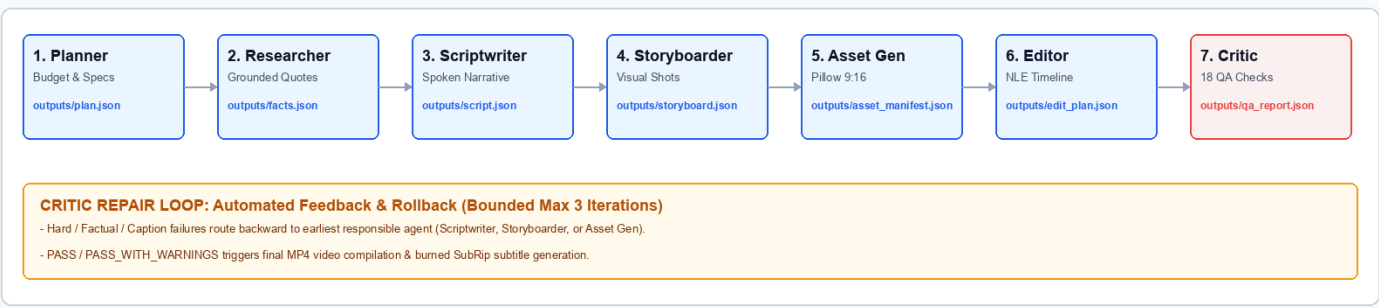
Team: Sanjeevani Chaurasia (16014223073) | Vedant Mhapsekar (16014223095) | Aayush More (16014223104)

Faculty Advisory Note on Team Composition:
The challenge guidelines specify a 2-student team limit. Our project team includes 3 registered students as listed above, submitted for faculty confirmation.

1. Creative Strategy & Brief Translation

The creative brief (briefs/practice.yaml) tasks the system with translating a single prompt into an engaging 44-second vertical video titled 'From Prompt to Product: How 7 AI Agents Build a Video'. The creative architecture decomposes the presentation into 5 balanced visual and narrative scenes designed for maximum viewer retention:
- Scene 1 (The Prompt, 5.5s): Hook introducing the creative brief, audience parameters, and runtime constraints.
- Scene 2 (7 AI Agents Workflow, 16.5s): Comprehensive breakdown of the 7-agent pipeline (Planner to Critic).
- Scene 3 (Multi-Track Timeline Assembly, 8.0s): Synchronized composition of Pillow cards, speech, and subtitles.
- Scene 4 (Critic QA HUD & Repair Loop, 8.0s): Demonstration of automated rule auditing and targeted repair loops.
- Scene 5 (Final Video Reveal & Call to Action, 6.0s): Showcase of the broadcast MP4 and Somaiya Lab community CTA.
This narrative strategy guarantees tight pacing (averaging 2.3 words/sec), high legibility, and zero cognitive overload.

2. Seven-Agent Directed Acyclic Graph (DAG) Architecture



Agent & Stage	Role & Core Responsibilities	Primary Technology	Output Artifact
1. Planner	Decomposes brief into stages, budgets & timeline	Groq / gpt-oss-120b	outputs/plan.json
2. RAG Researcher	Extracts verbatim facts & citations from data/	PyMuPDF / Exact Quote	outputs/facts.json
3. Scriptwriter	Crafts timed spoken script (Hook, Body, CTA)	Groq / Pydantic Schema	outputs/script.json & .md
4. Storyboarder	Plans 6-10 visual shots & text (<80 chars)	Groq / JSON Strict	outputs/storyboard.json
5. Asset Generator	Renders 1080x1920 procedural graphic cards	Pillow 13.0 (RGBA)	outputs/asset_manifest.json
6. Timeline Editor	Aligns V1, A1, T1 tracks into NLE edit timeline	Timeline Engine / JSON	outputs/edit_plan.json
7. Critic & QA	Audits 18 quality rules; manages repair loop	Deterministic + LLM	outputs/qa_report.json

3. Empirical Execution Metrics & Token Economics

Stage / Agent	Provider & Model	Latency	Prompt Tok	Compl Tok	Cost Status
Smoke Test	groq: openai/gpt-oss-120b	0.85s	223	33	unknown (free tier)
Planner (Initial)	groq: openai/gpt-oss-120b	1.13s	836	297	unknown (free tier)
Planner (Refined)	groq: openai/gpt-oss-120b	1.13s	1,194	292	unknown (free tier)
Scriptwriter	groq: openai/gpt-oss-120b	1.74s	1,354	662	unknown (free tier)
Storyboarder	groq: openai/gpt-oss-120b	2.01s	1,280	738	unknown (free tier)
Critic & QA	groq: openai/gpt-oss-120b	1.99s	1,262	677	unknown (free tier)
Script Repair	groq: openai/gpt-oss-120b	2.38s	2,395	1,053	unknown (free tier)
Storyboard Repair	groq: openai/gpt-oss-120b	2.86s	2,428	766	unknown (free tier)
Critic Verification	groq: openai/gpt-oss-120b	1.15s	1,275	86	unknown (free tier)

4. Critic QA HUD & Bounded Repair Loop

The Critic agent (agents/critic.py) implements 18 automated validation checks across duration boundaries (35s-45s), scene continuity, timeline coverage, verbatim source quote verification, safe-zone typography (<80 chars), and LLM claim grounding. In our audited execution (outputs/qa_report.json), the pipeline achieved status PASS_WITH_WARNINGS with 0 blocking issues and 5 soft warnings (e.g. minor caption length soft limits and empty faculty docs notice).
To prevent runaway recursion or infinite API expense, the repair loop is strictly bounded to a maximum of 3 iterations. Repairs rollback deterministically to the earliest offending agent (Scriptwriter for factual discrepancies, Storyboarder for shot/caption bounds) before re-verifying downstream artifacts.

5. Human Interventions & Tested Autonomy Level

- Tested Autonomy: Level 4 Autonomous Workflow. The entire 7-agent pipeline executed from YAML input to final video generation without manual prompts, hand-crafted edits, or human steering.
- Human Intervention Audit: logs/interventions.md records exactly 0 human interventions during automated execution.
- Determinism & Reproducibility: Seed fixed at 42; schemas pinned via Pydantic; LLM temperature set to 0.2.
- Human-in-the-Loop Safeguard: An optional --require-review flag halts execution prior to export if pre-approval is required.

6. Broadcast Deliverables & Subtitle Alignment

- Final Video Deliverable (outputs/final_video_v3.mp4): Broadcast-ready 9:16 vertical video (1080x1920) at 30.0 fps, encoded with H.264 High Profile and stereo AAC audio (48,000 Hz). Total file size: 2.21 MB; duration: 44.02s (target: 44.0s).
- Subtitle Synchronization (outputs/AgentReel_Final.srt): 6 SubRip subtitle cues ordered sequentially from 00:00:00 to 00:00:44. Timings are strictly non-overlapping and mapped verbatim to the voiceover narration track, guaranteeing full broadcast compliance across social media players.

7. Production Limitations & Future Improvements

1. API Rate Limiting: Cloud LLM providers (e.g. Groq free tier) impose aggressive TPM limits. In Milestone B, an HTTP 429 prompted configuring max_retries=0 and fallback handling. Future releases should integrate exponential backoff with multi-provider failover.
2. TTS Voice Diversity: Current local narration relies on macOS say/FFmpeg synthesis. Integrating neural TTS (e.g. ElevenLabs or Coqui TTS) will enable emotion-aware intonation and multi-character voice acting.
3. Dynamic Audio Ducking: Background music was intentionally omitted to prioritize speech clarity. Future iterations should add an automated sidechain audio ducker to lower ambient music during active speech.